

Track 2 — Senior / Tech Lead Mastery

This turns you from a Java developer into a production-ready tech-lead candidate. Every answer must carry a project example from WSI, CARB or payments. Run it on Saturdays plus 20 minutes on weekdays.

• Goal & Operating Model

- ☐ Explain maintainability, production readiness and design quality clearly.
- ☐ Build 5 short answers per week and **speak each twice**.
- ☐ Tie every concept to a real project (WSI / CARB / payments).

INTERVIEW-WEEK FLOOR

When Track 1 takes over for an interview, keep a single **20-minute Track 2 touch** for the week — one senior concept + one project example — so depth never drops to zero. Non-negotiable; everything else can pause.

• The 12-Week Senior / Lead Build

Week 1 — SOLID Principles

Explain SOLID with real project examples, not textbook definitions.

STUDY

- ☐ SRP
- ☐ OCP
- ☐ LSP
- ☐ ISP
- ☐ DIP

FRAME IT WITH

- ☐ Controller-service-repository separation
- ☐ Payment/fraud strategy extension without touching core flow
- ☐ Subclasses don't break parent behavior
- ☐ Small interfaces per client
- ☐ Spring DI on interfaces, not concretes

This week: Write 5 short answers and speak each twice. Signal: name where over-applying a principle would hurt.

Week 2 — 12-Factor App

Explain cloud-native Spring Boot service design.

STUDY

- ☐ Codebase, dependencies, config
- ☐ Backing services, build/release/run
- ☐ Processes, port binding, concurrency
- ☐ Disposability, dev/prod parity
- ☐ Logs, admin processes

FRAME IT WITH

- ☐ Profiles + env vars + secrets per environment
- ☐ Dockerized Spring Boot service
- ☐ Stateless REST APIs
- ☐ Actuator health checks
- ☐ CI/CD separates build / release / run

This week: Write 5 short answers and speak each twice.

Week 3 — Design Patterns

Explain patterns through real project examples.

STUDY

- ☐ Factory
- ☐ Strategy
- ☐ Builder
- ☐ Singleton
- ☐ Observer
- ☐ Decorator
- ☐ Template
- ☐ Adapter

FRAME IT WITH

- ☐ Strategy: payment/fraud rules
- ☐ Factory: client/processor by type
- ☐ Builder: Accertify/device payload
- ☐ Singleton: Spring beans
- ☐ Observer: Kafka event flow
- ☐ Adapter: external-client wrapper

This week: Write 5 short answers and speak each twice. Signal: when a simpler alternative beats the pattern.

Week 4 — API Design & Clean Architecture

Design clean, maintainable REST services.

STUDY

- ☐ DTOs vs entities
- ☐ Validation, error model
- ☐ Versioning, pagination
- ☐ Idempotency keys, correlation IDs
- ☐ API contracts, backward compatibility

FRAME IT WITH

- ☐ Cart/checkout API design
- ☐ Global exception model
- ☐ Idempotent payment endpoint
- ☐ DTO mapping hides entity internals
- ☐ Correlation ID across logs

This week: Write 5 short answers and speak each twice.

Week 5 — DDD-lite / Domain Modeling

Model the domain, not just tables.

STUDY

- ☐ Bounded context
- ☐ Aggregate, value object
- ☐ Domain service, repository
- ☐ Anti-corruption layer

FRAME IT WITH

- ☐ Cart, checkout, payment, fraud/risk, order boundaries
- ☐ Aggregates around invariants
- ☐ ACL around legacy/3rd-party

This week: Write 5 short answers and speak each twice. Signal: where a wrong boundary caused coupling.

Week 6 — Observability & Production Readiness

Talk like someone who supports production.

STUDY

- ☐ Structured logs, correlation ID
- ☐ Metrics, traces
- ☐ Health checks, alerts, dashboards
- ☐ Runbooks, incident triage

FRAME IT WITH

- ☐ Checkout latency, Kafka lag
- ☐ Accertify downstream errors
- ☐ CloudWatch / ELK / Splunk
- ☐ Runbook for downstream failure

This week: Write 5 short answers and speak each twice.

Week 7 — Testing Strategy

Explain how you ensure quality as a lead.

STUDY

- ☐ Unit, Mockito
- ☐ Integration, TestContainers
- ☐ Contract, API, performance
- ☐ Coverage and review gates

This week: Write 5 short answers and speak each twice.

FRAME IT WITH

- ☐ JUnit/Mockito service tests
- ☐ Kafka integration test
- ☐ Sonar/Jacoco gates
- ☐ Code-review checklist

Week 8 — CI/CD & DevOps

Explain deployment flow and rollback.

STUDY

- ☐ Jenkins, Maven/Gradle
- ☐ Docker, K8s, probes
- ☐ Secrets
- ☐ Blue-green / canary, rollback

This week: Write 5 short answers and speak each twice.

FRAME IT WITH

- ☐ Pipeline stages
- ☐ Readiness/liveness probes
- ☐ Release validation
- ☐ Rollback strategy

Week 9 — Security

Explain secure backend design.

STUDY

- ☐ AuthN / AuthZ
- ☐ JWT, OAuth2, OIDC, RBAC
- ☐ OWASP, SQL-injection
- ☐ Secrets management

This week: Write 5 short answers and speak each twice.

FRAME IT WITH

- ☐ Cognito/JWT in CARB
- ☐ JWT filter chain
- ☐ RBAC by role/claim
- ☐ Never log secrets/tokens

Week 10 — Performance & Scalability

Find bottlenecks and scale safely.

STUDY

- ☐ Profiling, indexes
- ☐ Cache-aside, connection/thread pools
- ☐ Async, backpressure
- ☐ Kafka scaling, load testing

This week: Write 5 short answers and speak each twice.

FRAME IT WITH

- ☐ Checkout latency triage
- ☐ HikariCP pool
- ☐ Kafka partitions/consumers
- ☐ JMeter baseline

Week 11 — Leadership & Delivery

Prepare lead-level stories and behaviors.

STUDY

- ☐ Mentoring, reviews, estimation
- ☐ Conflict, incidents
- ☐ Tech debt, stakeholder comms
- ☐ Decision records

FRAME IT WITH

- ☐ Tech Anchor at WSI
- ☐ Offshore coordination
- ☐ Driving a decision
- ☐ Production incident ownership

This week: Write 5 lead-level stories and speak each twice.

Week 12 — Architecture Review Board Mode

Drive design decisions across teams.

STUDY

- ☐ Decision records (ADRs)
- ☐ Trade-off docs, standards
- ☐ Governance
- ☐ Migration roadmap

FRAME IT WITH

- ☐ How you drove a cross-team decision
- ☐ A migration you sequenced
- ☐ Standards you set or enforced

This week: Write 3 decision-record style answers and speak each.

· Project Mapping Table

CONCEPT	PROJECT EXAMPLE
SRP	Controller-service-repository separation in checkout APIs.
OCP	Payment/fraud rules via Strategy so new rules don't change orchestration.
Builder	Accertify / device payload creation.
Adapter	Wrapper around an external fraud/risk client.
Observer	Kafka event flow for risk / eGift / gift-card events.
12-factor config	Profiles / env vars / secrets per environment.
Idempotency	Payment/checkout/Kafka processing on a business key / event ID.
Observability	Correlation ID from API to downstream logs; Kafka lag dashboard.
Security	Cognito/OAuth/JWT in CARB; JWT filter chain in Spring Security.
Leadership	Tech Anchor: reviews, offshore coordination, decisions, delivery.

· Senior / Tech-Lead Question Bank

SOLID / Patterns

- 1

Explain OCP with a real example.
- 2

Where did you use Strategy?
- 3

Factory vs Strategy.
- 4

How is Builder useful for payloads?
- 5

Spring singleton beans vs the Singleton pattern.
- 6

Adapter for third-party integration.
- 7

How do you avoid over-engineering patterns?
- 8

How do you apply DIP in Spring?
- 9

Risk of violating SRP.
- 10

How do you review code for SOLID?

Architecture / API

- 1 How do you design a REST API?
- 2 DTO vs entity.
- 3 Error response format.
- 4 API versioning strategy.
- 5 Idempotency-key design.
- 6 Correlation-ID propagation.
- 7 Pagination and sorting.
- 8 Backward compatibility.
- 9 How do you split services?
- 10 How do you document API contracts?

Production

- 1 How do you debug production issues?
- 2 What logs do you add?
- 3 Metrics you track for checkout.
- 4 How do you monitor Kafka lag?
- 5 How do you design alerts?
- 6 Runbook for a downstream failure.
- 7 Incident communication.
- 8 How do you validate a release?
- 9 A good rollback plan.
- 10 How do you handle tech debt?

Testing / Quality

- 1 Unit vs integration tests.
- 2 Mockito best practices.
- 3 TestContainers usage.
- 4 Contract testing.
- 5 Kafka testing strategy.
- 6 API testing.
- 7 Performance testing.
- 8 Sonar/Jacoco gates.
- 9 Code-review checklist.
- 10 Mentoring juniors on testing.

Leadership

- 1 How do you lead offshore teams?
- 2 How do you handle disagreement?
- 3 How do you estimate work?
- 4 How do you split tasks?
- 5 How do you mentor?
- 6 How do you run code reviews?
- 7 How do you make design decisions?
- 8 How do you handle missed deadlines?
- 9 How do you communicate risk?
- 10 Why hire you as a lead?

• Answer Templates

TOPIC	TEMPLATE
SOLID	Definition → project example → benefit → trade-off → how you enforced it in reviews.
12-factor	Principle → Spring Boot implementation → production benefit → mistake to avoid.
Design pattern	Problem → pattern → implementation → why not a simpler alternative → project usage.
Observability	Logs + metrics + traces + correlation ID + alert + runbook + a project incident.
Leadership	Situation → decision → collaboration → measurable outcome → lesson learned.

• Production Debugging Scenarios

The format senior/lead interviews actually use. Don't jump to a fix — walk the method out loud, then narrow to the likely layer.

DIAGNOSTIC LADDER

Reproduce / confirm the symptom → look at **metrics and dashboards** first → narrow the layer (app code, GC, threads, DB, downstream, network) → form a hypothesis → verify with **logs / traces / heap or thread dumps** → fix → add a guardrail (alert, test, runbook) so it cannot silently recur.

- 1 API latency jumped 200ms → 5s — how do you debug?
- 2 JVM memory grows continuously — find the leak
- 3 CPU pinned at 100% — what do you check?
- 4 Database slow right after a deploy
- 5 Thread-pool queue keeps growing
- 6 Kafka consumer lag climbing — how do you fix it?
- 7 Checkout p95 regressed overnight
- 8 A downstream dependency started timing out
- 9 Design a cache for 10M users
- 10 Design a rate limiter with Redis

• Track 2 Coverage Map

- ☐ Explain all SOLID principles with project examples.
- ☐ Explain the 12 factors in Spring Boot terms.
- ☐ Map Strategy / Factory / Builder / Adapter to a real project.
- ☐ Design a clean REST API: DTOs, validation, errors, versioning, idempotency.
- ☐ Explain logs, metrics, traces, correlation IDs and dashboards.
- ☐ Explain testing: unit, integration, Kafka/API and quality gates.
- ☐ Explain CI/CD, Docker, K8s, probes, rollback and release validation.
- ☐ Explain JWT / OAuth / OIDC / RBAC and secure API design.
- ☐ Explain domain boundaries and where a wrong one caused coupling.
- ☐ Tell 5 lead-level behavioral stories and drive one design decision.

READINESS

These boxes mean you have *covered* the topic. The gauge is whether you can deliver each in a mock — score them on the Master's Mock Scorecard.